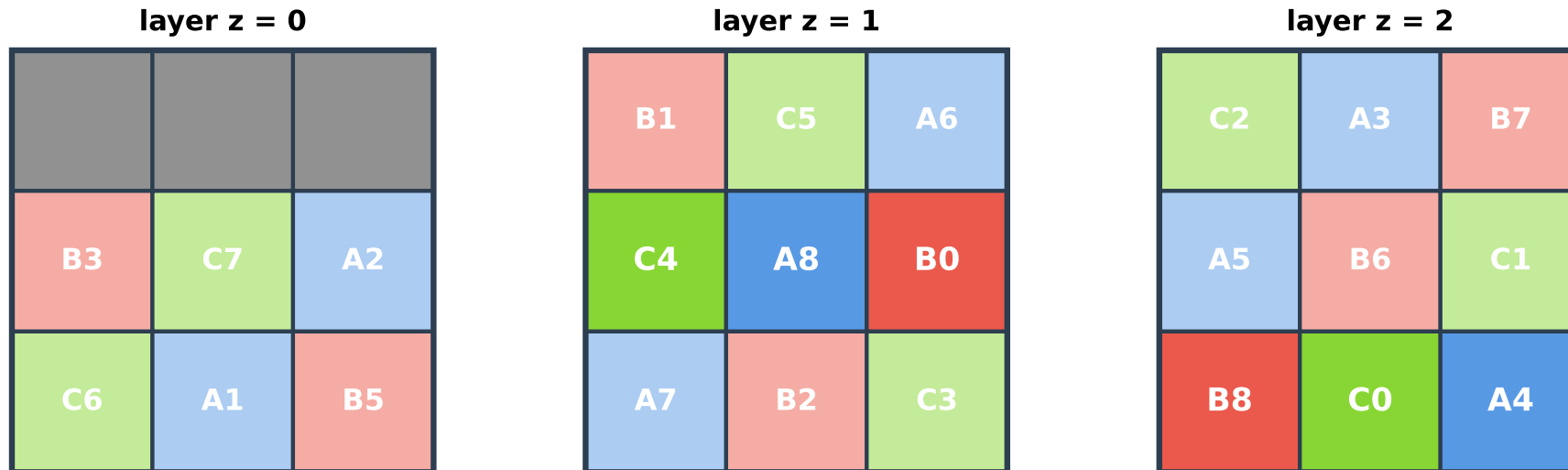


Recovery of a Lost Rod and a Lost Slab in the 3D DR Code

explicit DR-recoverable template of Section 4.2; erased blocks are shown in grey, the surviving partners of each erased block in full colour

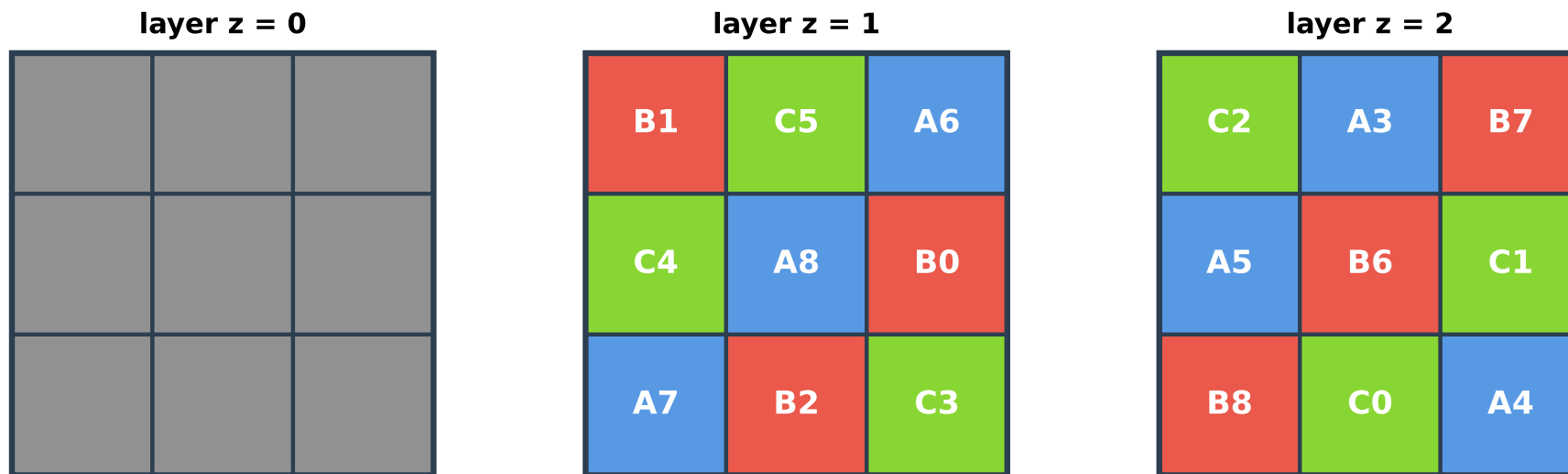
(a) Rod loss: the three cells (x, 0, 0) of layer z = 0 are erased (blocks A0, B4, C8)



$$a_0 = b_0 \oplus c_0 \quad b_4 = a_4 \oplus c_4 \quad c_8 = a_8 \oplus b_8$$

each erased block is the only lost member of its parity triple, so its two surviving partners restore it by one XOR

(b) Slab loss: the whole layer z = 0 is erased (nine blocks, parity indices 0-8 once each)



$$a_0 = b_0 \oplus c_0, \quad b_1 = a_1 \oplus c_1, \quad \dots, \quad b_5 = a_5 \oplus c_5$$

every parity triple loses exactly one member, so all nine blocks are restored from their partners in layers z = 1 and z = 2